

Credit Risk & Pricing Strategy Analysis

(LendingClub Loan Portfolio)

1. Objective

This analysis evaluates credit risk, loan performance and risk-based pricing across a LendingClub loan portfolio of approximately **2.26 million loans**.

The analysis focuses on four questions:

How does observed credit risk vary across borrower risk grades?

How does loan term relate to observed credit performance?

Does pricing increase as observed risk increases?

Which sub-grade transitions warrant further pricing investigation?

2. Key Findings

1. Credit risk increases substantially across risk grades

Observed Credit Loss / Default rates increase materially from Grade A to Grade G, rising from approximately **3.3% for Grade A to 38.1% for Grade G**.

Average interest rates also increase across the risk spectrum, from approximately **7.1% for Grade A to 28.1% for Grade G**.

Interpretation: The results indicate strong risk differentiation across LendingClub grades, with substantially higher observed credit risk concentrated in Grades E–G.

2. Loan-level default rate and exposure rate provide different views of risk

Approximately **11.91% of loans** are classified as Credit Loss / Default. These loans represent approximately **12.31% of original loan exposure**, equivalent to approximately **\$4.19 billion**.

The difference between the two measures reflects variation in loan sizes.

Interpretation: The Credit Loss / Default Rate measures the **frequency of affected loans**, while the Default Exposure Rate measures the **share of original loan dollars associated with those loans**. The \$4.19 billion figure represents defaulted loan exposure, not realised credit loss, because recoveries and loss severity are not incorporated.

3. Loan-term risk varies by borrower risk segment

At portfolio level, 60-month loans show a higher observed Credit Loss / Default rate than 36-month loans, at approximately **16.2% versus 10.2%**. Average interest rates are also higher for 60-month loans.

However, the relationship changes when analysed within individual grades. Grades A–C show lower observed loss rates for 60-month loans than their 36-month counterparts, while Grades D–G generally show the opposite pattern.

Interpretation: Loan term should therefore be evaluated alongside borrower risk rather than in isolation. The observed differences may reflect borrower and loan composition rather than an independent causal effect of term.

4. Pricing broadly increases with observed risk

At sub-grade level, average interest rates generally increase alongside observed Credit Loss / Default rates.

For example, **A1** has an average interest rate of approximately **5.6%** and an observed Credit Loss / Default rate of approximately **1.6%**, while higher-risk sub-grades such as **G3** have an average interest rate above **28%** and an observed Credit Loss / Default rate of approximately **40%**.

Interpretation: The broad upward movement in pricing and observed risk indicates substantial risk-based pricing differentiation. However, individual sub-grade movements are not perfectly monotonic and should be investigated rather than automatically interpreted as pricing errors.

5. Selected sub-grade transitions warrant pricing investigation

To identify potentially unusual pricing movements, each sub-grade was compared with the preceding sub-grade using changes in **average interest rate** and **default exposure rate**.

The analysis identifies:

Under-pricing review: D1, E1, F1, F2, F4 and G2

Potential over-pricing review: E5 and G1

For example, the transition from **D5 to E1** shows approximately a **5.37 percentage-point increase in default exposure** compared with only a **0.27 percentage-point increase in average interest rate**.

Interpretation: These flags identify areas for further investigation. They do **not** establish that a sub-grade is incorrectly priced. Differences may reflect borrower characteristics, loan characteristics, origination vintage, underwriting policy or other portfolio factors.

3. Recommendations

Strengthen granular risk monitoring

Continue monitoring risk at grade and sub-grade level, particularly Grades E–G, where observed Credit Loss / Default rates and default exposure are substantially higher.

Review pricing at sub-grade level

Use the identified pricing transitions as a starting point for deeper pricing investigation rather than relying solely on broad grade-level averages.

Incorporate loan term into risk segmentation

Evaluate loan term alongside borrower risk because the relationship between term and observed credit performance differs across grades.

Monitor both risk frequency and exposure

Track both the proportion of loans entering Credit Loss / Default and the original loan dollars associated with those loans to understand both **risk frequency and financial exposure**.

4. Methodology & Limitations

Loan outcomes were derived from loan status and grouped into **Performing, Delinquent / At Risk, Credit Loss / Default and Unknown** categories.

Credit Loss / Default Rate measures the proportion of loans classified as Credit Loss / Default. Default Exposure Rate measures the proportion of original loan amount associated with those loans. Exposure is therefore not equivalent to realised credit loss.

Pricing alignment flags use analyst-defined screening rules based on changes between adjacent sub-grades:

Under-pricing review: Default Exposure Change ≥ 2 percentage points and Interest Rate Change < 1 percentage point.

Potential over-pricing review: Default Exposure Change ≤ -1 percentage point and Interest Rate Change ≥ 0.5 percentage points.

These flags are screening indicators rather than confirmed pricing errors. The analysis is primarily descriptive and does not establish causal relationships.

A more advanced analysis could incorporate **origination vintage, borrower characteristics, loan characteristics, underwriting variables and recovery data** to evaluate expected loss and pricing more robustly.

5. Conclusion

The analysis demonstrates a strong relationship between borrower risk segmentation, observed credit losses and loan pricing across the LendingClub portfolio.

Higher-risk grades and sub-grades exhibit substantially higher observed Credit Loss / Default rates and default exposure, while pricing generally increases with risk. However, the Grade $\times$ Term and sub-grade analyses show that portfolio-level relationships can conceal important differences within individual risk segments.

The pricing alignment framework provides a practical way to identify areas for further investigation while recognising that observed differences do not, on their own, establish pricing errors.